

VARSHINI K

DATA ANALYST | SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

- **Data Analyst and Data Scientist** with **2 years of experience** in **data analyst, machine learning, and cybersecurity**, reporting, ETL pipeline development, and supply-chain focused analysis.
- Proficient in **Python, Java, SQL, Flask, React.js, Django, and Node.js**, Power BI, Visual Studio, Excel, and dashboard development for operational decision-making.
- Hands-on experience in requirements gathering, **data cleaning, data migration, strategic planning, business process improvement, forecasting**, and inventory management.
- Experienced in designing and developing ETL **workflows, data ingestion pipelines, RESTful APIs**, and visualization dashboards for real-time insights.
- Experienced in API development using Flask, database systems, and real-time data processing using **Spark, Kafka, and cloud platforms (AWS, Azure, GCP)**.
- Passionate about building **scalable, secure, and high-quality data solutions** that support **analytics and machine learning initiatives**.
- Prior experience at TCS in **cybersecurity, DLP monitoring, access management, and ServiceNow dashboards**, ensuring **data protection and HIPAA/GDPR compliance**.
- Applied **machine learning and explainable AI** to projects including **fire detection, healthcare prediction, and resume automation**, integrating ML into production-ready websites, passionate about solving complex infrastructure challenges.
- Strong collaborator with hands-on expertise in **Agile, CI/CD, and DevOps (Azure DevOps, GitHub Actions, Terraform, Airflow)**, ensuring reliable, efficient, and secure data-driven applications.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, Java, JavaScript, TypeScript, C++, C#.
- **Web Development:** Flask, Node.js, ReactJS, HTML5, CSS3, Django, Responsive Designs.
- **Data Infrastructure & ETL:** Apache Kafka, Spark, MongoDB Atlas, Cloudfare, Big Data, SQL, Postgres, RDBMS, Cassandra, Data Pipelines, API Design, RESTful APIs.
- **Visualization & Monitoring:** Data Preprocessing, Grafana, Matplotlib, Seaborn, Heatmaps, Scatter/Count/Bar/Pie Charts), Algorithmic Optimization, OOPs, Dashboards,
- **Machine Learning & AI:** ML Models, Evaluation Metrics (Accuracy, Precision, Recall, F1-score, Confusion Matrix), SHAP (Shapley Values, Summary/Force/Dependency/Waterfall Plots).
- **Cloud & DevOps:** AWS, Azure, GCP, Terraform, GitHub Actions, Azure DevOps, GitLab CI/CD, Google meet.
- **Supply Chain & Business Operations:** Inventory Management, Logistics Processes, Forecasting, Strategic Planning, Requirements Gathering, Risk Mitigation, Business Process Improvement

- **Security & ITSM:** DLP Tools (Symantec, Microsoft Purview), ServiceNow, Access Management & Authentication (Session-based Auth, IAM, Role-based Access), Cybersecurity & Compliance
- **Development Tools:** Unix/Linux, Windows, Visual Studio.

WORK EXPERIENCE

Kent State University

Kent, Ohio

Role: Data Analyst Intern

Aug 2024 – Dec 2024

- Designed and developed analytical solutions using **Python and SQL** to identify trends and generate actionable business insights from large datasets.
- Built **Power BI dashboards** and reports to communicate KPIs, forecasts, and operational metrics to non-technical stakeholders.
- Developed scalable **ETL pipelines** using Kafka, Spark, and Snowflake, improving data processing efficiency and reliability.
- Collaborated with cross-functional teams to define project goals, incorporate business context, and deliver data-driven recommendations.
- Implemented data validation and quality checks, reducing reporting errors and improving decision accuracy.
- Documented analytical workflows and reusable components to support future analytics initiatives.

Tata Consultancy Services

Hyderabad, India

Role: Associate Data Analyst

July 2022 – July 2023

- Conducted data analysis on DLP events and security workflows ensuring accuracy, compliance, and incident resolution.
- Utilized **SQL, ServiceNow dashboards**, and KPI metrics to support operational analytics and trend monitoring.
- Supported **logistics and asset-related workflows** by tracking access provisioning, schedules, and workflow performance.
- Worked on **requirements gathering**, documentation, and workflow assessments for cybersecurity operations. Identified gaps and executed **business process improvements**, reducing manual efforts and improving response time.
- Applied **strategic planning** for dashboard enhancements and monitoring process optimization. • Used analytical insights to support **risk mitigation** and supply-chain related data handling in secure environments. Participated in automation initiatives to streamline incident-response processes and operational workflows.

Tata Consultancy Services

Tamilnadu, India

Role: Intern Machine Learning Engineer Developer

Jan 2022- July 2022

- Developed and deployed **machine learning models (Linear Regression, Logistic Regression, Decision Tree, Random Forest)** on the Titanic dataset to predict survival outcomes with high accuracy.
- Performed **data preprocessing**, feature engineering, and model development using **Python, NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn**.
- Conducted requirements gathering for **ML model development** and ensured alignment with analytical objectives.
- Built and evaluated models using metrics including **Accuracy, Precision, Recall, F1-score, and Confusion Matrix**. Implemented **SHAP-based explainable AI techniques**, generating force, summary,

dependency, and waterfall plots to interpret model behavior.

- Designed and deployed a **Django-based application to display SHAP visualizations** and ML outputs through interactive dashboards and visualizations to support forecasting, strategic planning, and decision-making. Optimized data-processing workflows, improving performance and contributing to measurable process-efficiency gains.
- Collaborated with team mentors to deliver an end-to-end ML solution combining explainable AI and full-stack web technologies.

PROJECTS

ProResume – Capstone Project (Full Stack Resume Builder)

Environment: ReactJS | Flask | Mango DB | HTML | CSS | Java | JavaScript

- Collaborated on a full-stack web application to generate professional resumes, leveraging core computer science concepts such as **object-oriented design**, **data structures**, and **algorithmic logic**.
- Contributed to the resume builder module using **Java-based data validation** logic and dynamic form management, ensuring seamless **user input handling and template generation**.
- Integrated real-time preview, **CRUD** operations, and PDF export features, demonstrating hands-on experience with **RESTful APIs**, **user authentication**, and **full lifecycle web development**.
- Collaborated with a 6-member team using **GitHub** and **Google Meet**, driving iterative **development**, **bug resolution**, and **UI responsiveness** across devices.
- Applied optimization techniques to enhance resume generation speed and reduce form processing lag by **35%**, **showcasing attention to algorithmic performance**.
- **Applied analytical thinking to optimize system performance and user workflows.**

Hospital Management System — Java, Applet, DBMS, HTML, CSS

- Designed and implemented a **Java-based hospital management system** using **object-oriented programming (OOP)** principles to ensure modularity, scalability, and maintainability.
- Developed the user interface using **Java Applets**, **HTML**, and **CSS**, creating an interactive and responsive front-end experience for healthcare providers and patients.
- Integrated the application with a **SQL-based DBMS**, enabling secure data storage, efficient query execution, and real-time access to patient records.
- Applied **algorithmic optimization** techniques to improve system responsiveness and reduce scheduling conflicts by **25%**. Demonstrated proficiency in Java through hands-on development of a full front-end logic layer integrated with a database backend.
- Developed predictive models and explainable AI visualizations to support data-driven decision-making. Created reusable modeling components and documentation to enable future analytics projects.

EDUCATION

Kent State University

2023-2025

MS in computer science | 3.8 CGPA